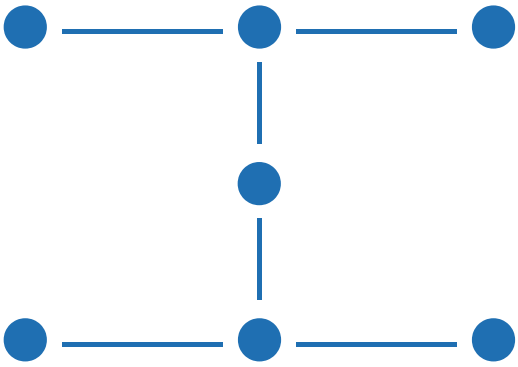
IT SKILLS

MULTI-SITE ENTERPRISE NETWORK

Network Design, Implementation
and Verification Documentation

GERMANY HQ • EGYPT • UNITED KINGDOM
INDIA • Saudi Arabia • DUBAI



PREPARED BY

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Cisco Packet Tracer | Version 1.0 | October 2026

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1. Executive Summary

IT Skills is a fictional multinational company with its headquarters in Germany and five branch offices in Egypt, the United Kingdom, India, Saudi Arabia (KSA) and Dubai. This project designs, implements and verifies a routed enterprise network that connects all six sites over point-to-point WAN links, segments every site with VLANs, and centralizes name resolution, web hosting and file transfer services at headquarters.

Key outcomes:

- **Dynamic routing:** OSPF (process 1, area 0) exchanges routes between the HQ edge router and all five branch edge routers, and between each edge router and its local core switch.
- **Segmentation:** nine VLANs with identical IDs at every site, routed locally by a multilayer core switch.
- **Centralized services:** DNS, web and TFTP servers at HQ in VLAN 70. The website www.itskills.com resolves and loads from branch sites.
- **Guest isolation:** an extended ACL on each Guest VLAN blocks private (RFC 1918) destinations while permitting DNS and web access to the central servers.
- **Verified operation:** GER-EDGE has six OSPF neighbors (five branch edge routers and GER-CORE), all in FULL state, and cross-branch, DNS and web tests succeed with 0% packet loss.

2. Scope and Objectives

Objectives:

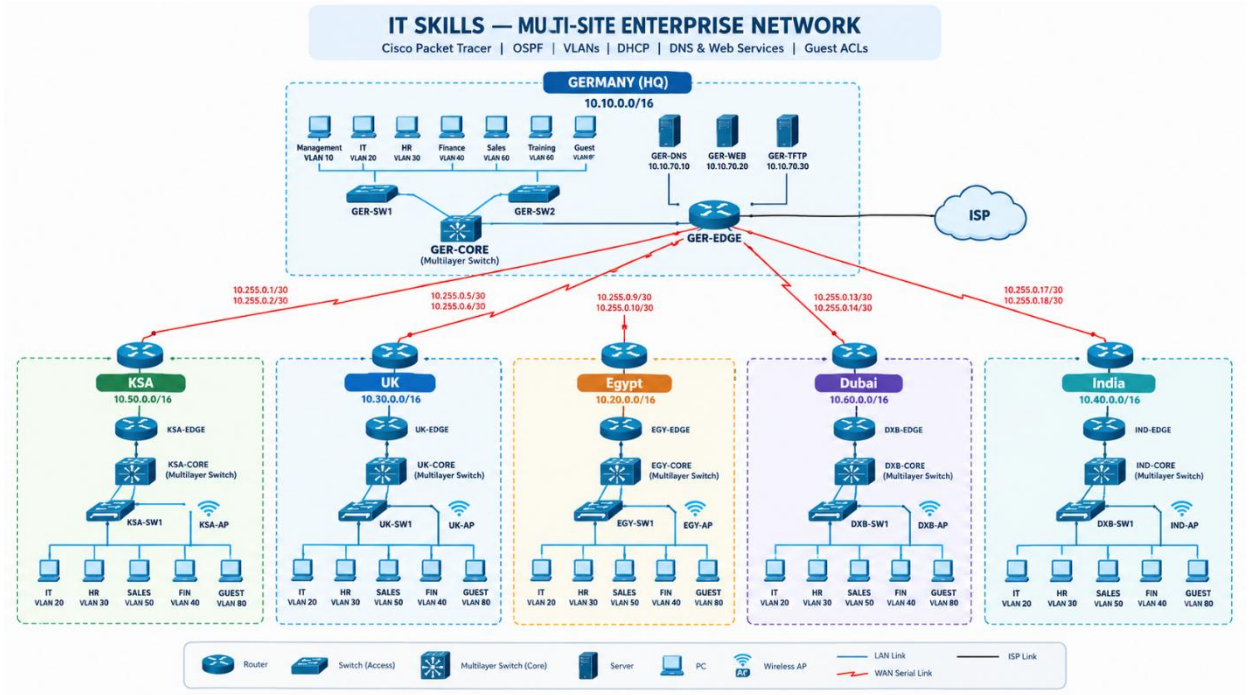
1. Connect six sites through routed WAN links and exchange routes dynamically with OSPF area 0.
2. Separate departments, servers and guest devices into VLANs, with inter-VLAN routing on each site's multilayer core switch.
3. Provide client addressing through DHCP and central name resolution through the HQ DNS server.
4. Host the company website and a TFTP service at HQ.
5. Prevent Guest VLAN clients from reaching internal private networks while preserving their access to the approved central services (DNS and web).

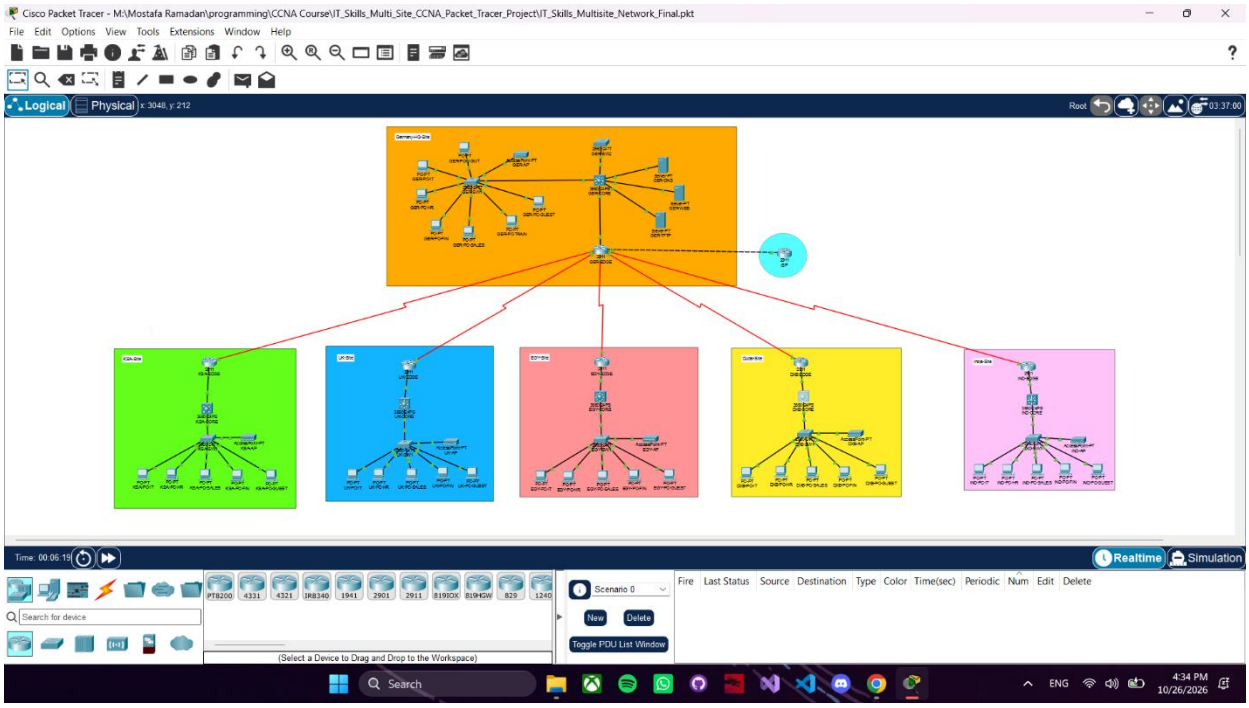
The project is a Cisco Packet Tracer simulation. ISP-side routing and Internet access are outside the validated test scope.

3. Network Architecture

3.1 Topology overview

The WAN is a hub-and-spoke design with Germany HQ as the hub. Each branch edge router connects to GER-EDGE over a dedicated serial /30 link, so inter-branch traffic transits headquarters. GER-EDGE also has an upstream link to the ISP router.





Inside every site the network follows a three-layer structure:

- **Access layer:** user PCs and a wireless access point connect to an access switch (SW1). Germany HQ uses two access switches, GER-SW1 and GER-SW2.

- **Core layer:** a multilayer switch hosts the VLAN interfaces (SVIs), performs inter-VLAN routing, serves DHCP and runs OSPF.
- **Edge layer:** a router connects the site to the WAN. The edge and core are joined by a routed transit subnet (10.<site>.254.0/30).

3.2 Site summary

Site	Address block	Edge router	Core switch	Access switch
Germany HQ	10.10.0.0/16	GER-EDGE	GER-CORE	GER-SW1, GER-SW2
Egypt	10.20.0.0/16	EGY-EDGE	EGY-CORE	EGY-SW1
United Kingdom	10.30.0.0/16	UK-EDGE	UK-CORE	UK-SW1
India	10.40.0.0/16	IND-EDGE	IND-CORE	IND-SW1
KSA	10.50.0.0/16	KSA-EDGE	KSA-CORE	KSA-SW1
Dubai	10.60.0.0/16	DXB-EDGE	DXB-CORE	DXB-SW1

3.3 Equipment

Role	Platform	Devices
Edge routers	Cisco 2911	GER-EDGE, EGY-EDGE, UK-EDGE, IND-EDGE, KSA-EDGE, DXB-EDGE
ISP router	Cisco 2911	ISP
Core switches (Layer 3)	Catalyst 3560-24PS	One per site
Access switches (Layer 2)	Catalyst 2960-24TT	GER-SW1, GER-SW2 and one per branch
Wireless access points	Access Point-PT	One per site (for example GER-AP, KSA-AP)
Servers	Server-PT	GER-DNS, GER-WEB, GER-TFTP
End devices	PC-PT	Named SITE-PC-DEPARTMENT (for example KSA-PC-IT)

3.4 Centralized services at Germany HQ

Device	IPv4 address	Role
GER-DNS	10.10.70.10/24	Central DNS. www.itskills.com resolves to 10.10.70.20
GER-WEB	10.10.70.20/24	Company website, tested over HTTP. The Guest ACL also permits HTTPS (TCP 443)
GER-TFTP	10.10.70.30/24	TFTP service

The servers use static addressing in VLAN 70. Their default gateway is 10.10.70.1, the VLAN 70 SVI on GER-CORE.

4. IP Addressing and VLAN Design

4.1 Addressing conventions

- **Site block:** 10.<site>.0.0/16, where the second octet is 10 (Germany), 20 (Egypt), 30 (UK), 40 (India), 50 (KSA) and 60 (Dubai).
- **VLAN subnets:** 10.<site>.<VLAN ID>.0/24, with the core SVI as the default gateway at .1. Germany VLAN 20 is 10.10.20.0/24 (gateway 10.10.20.1). Dubai VLAN 20 is 10.60.20.0/24 (gateway 10.60.20.1).
- **Edge-to-core transit:** 10.<site>.254.0/30, with the core uplink ending in .2 (for example 10.20.254.2 on EGY-CORE Gi0/1).
- **WAN links:** /30 subnets carved from 10.255.0.0.
- **ISP link:** 203.0.113.0/30 (GER-EDGE .2, ISP .1).

4.2 VLAN plan

VLAN IDs and names are identical at every site.

VLAN	Name	Purpose	Germany subnet
10	MANAGEMENT	Network management	10.10.10.0/24
20	IT	IT department	10.10.20.0/24
30	HR	Human Resources	10.10.30.0/24
40	FINANCE	Finance	10.10.40.0/24
50	SALES	Sales	10.10.50.0/24
60	TRAINING	Training users	10.10.60.0/24
70	SERVERS	Central servers (HQ)	10.10.70.0/24
80	GUEST	Guest devices	10.10.80.0/24
99	NATIVE-MGMT	Native VLAN on trunks	10.10.99.0/24

Branch sites host IT, HR, Sales, Finance and Guest endpoints. Germany HQ additionally hosts Management and Training endpoints and the central servers.

4.3 Site address plan

Site	Block	VLAN 20 (IT)	VLAN 80 (Guest)	Edge-to-core transit
Germany HQ	10.10.0.0/16	10.10.20.0/24	10.10.80.0/24	10.10.254.0/30
Egypt	10.20.0.0/16	10.20.20.0/24	10.20.80.0/24	10.20.254.0/30
United Kingdom	10.30.0.0/16	10.30.20.0/24	10.30.80.0/24	10.30.254.0/30
India	10.40.0.0/16	10.40.20.0/24	10.40.80.0/24	10.40.254.0/30
KSA	10.50.0.0/16	10.50.20.0/24	10.50.80.0/24	10.50.254.0/30

Site	Block	VLAN 20 (IT)	VLAN 80 (Guest)	Edge-to-core transit
Dubai	10.60.0.0/16	10.60.20.0/24	10.60.80.0/24	10.60.254.0/30

4.4 Trunking

Core-to-access uplinks are static IEEE 802.1Q trunks with native VLAN 99 and an explicit allowed list of VLANs 10, 20, 30, 40, 50, 60, 70, 80 and 99. VLAN 1 is not carried. On DXB-CORE, `show interfaces trunk` confirms Gig0/2 in mode on, encapsulation 802.1q, status trunking, native VLAN 99, with all nine VLANs allowed, active and forwarding.

4.5 DHCP

- DHCP pools run on each site's core switch for the client VLANs.
- Addresses .1 through .20 of each pool are excluded, reserving them for gateways and infrastructure. The first client lease is therefore .21. For example, the India IT PC holds 10.40.20.21.
- Clients use the central DNS server 10.10.70.10 for name resolution.
- Servers in VLAN 70 are statically addressed.

5. WAN Connectivity and OSPF

5.1 WAN links

Link	Subnet	GER-EDGE interface	Germany side	Branch side
Germany–Egypt	10.255.0.0/30	Serial0/0/0	10.255.0.1	10.255.0.2
Germany–UK	10.255.0.4/30	Serial0/0/1	10.255.0.5	10.255.0.6
Germany–India	10.255.0.8/30	Serial0/1/0	10.255.0.9	10.255.0.10
Germany–KSA	10.255.0.12/30	Serial0/1/1	10.255.0.13	10.255.0.14
Germany–Dubai	10.255.0.16/30	Serial0/2/0	10.255.0.17	10.255.0.18

The Germany–ISP link uses 203.0.113.0/30. GER-EDGE does not form an OSPF adjacency with the ISP router.

5.2 OSPF design

OSPF runs as a single area: process ID 1, area 0. Router IDs follow a fixed pattern so that every device is easy to identify in neighbor tables and logs.

Site	Edge router ID	Core router ID
Germany HQ	1.1.1.1	11.11.11.11
Egypt	2.2.2.2	12.12.12.12

Site	Edge router ID	Core router ID
United Kingdom	3.3.3.3	13.13.13.13
India	4.4.4.4	14.14.14.14
KSA	5.5.5.5	15.15.15.15
Dubai	6.6.6.6	16.16.16.16

Adjacencies: GER-EDGE peers with the five branch edge routers over the serial WAN links and with GER-CORE over its transit link. Each branch core switch peers with its local edge router. On the HQ edge-to-core segment, GER-CORE is the designated router (FULL/DR).

5.3 Verification: OSPF neighbors on GER-EDGE

Output of `show ip ospf neighbor`, all neighbors in FULL state:

Neighbor ID	Device	State	Neighbor address	Interface
2.2.2.2	EGY-EDGE	FULL/-	10.255.0.2	Serial0/0/0
11.11.11.11	GER-CORE	FULL/DR	10.10.254.2	GigabitEthernet0/1
3.3.3.3	UK-EDGE	FULL/-	10.255.0.6	Serial0/0/1
4.4.4.4	IND-EDGE	FULL/-	10.255.0.10	Serial0/1/0
6.6.6.6	DXB-EDGE	FULL/-	10.255.0.18	Serial0/2/0
5.5.5.5	KSA-EDGE	FULL/-	10.255.0.14	Serial0/1/1

The branch-side adjacency is confirmed on DXB-CORE, which logged neighbor 6.6.6.6 moving from LOADING to FULL on GigabitEthernet0/1.

6. Network Services

- **DNS:** GER-DNS (10.10.70.10) holds the record www.itskills.com pointing to 10.10.70.20. All sites query this single server across the WAN.
- **Web:** GER-WEB (10.10.70.20) serves the IT Skills home page. Browsing to http://www.itskills.com from EGY-PC-IT displays the page, which confirms both DNS resolution and HTTP delivery across the WAN. Only HTTP access was tested; HTTPS was not part of the verification.
- **TFTP:** GER-TFTP (10.10.70.30) provides a central TFTP service, for example for device configuration and file transfer.

Branch networks reach these services through OSPF-learned routes via GER-EDGE.

7. Guest VLAN Security

Each site applies an extended ACL named SITE-GUEST-IN (for example DXB-GUEST-IN) inbound on the VLAN 80 SVI of its core switch. The policy keeps guests away from internal private networks while retaining access to approved central services and non-private destinations.

Entry	Purpose	Action
1	ICMP to the local Guest gateway	Permit
2–3	DNS (UDP and TCP port 53) to 10.10.70.10	Permit
4–5	HTTP (TCP 80, tested) and HTTPS (TCP 443, permitted) to 10.10.70.20	Permit
6–8	10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16	Deny
9	All other destinations	Permit

Configured ACL on DXB-CORE (`show access-lists DXB-GUEST-IN`):

```
Extended IP access list DXB-GUEST-IN
    permit icmp any host 10.60.80.1
    permit udp any host 10.10.70.10 eq domain
    permit tcp any host 10.10.70.10 eq domain
    permit tcp any host 10.10.70.20 eq www
    permit tcp any host 10.10.70.20 eq 443
    deny ip any 10.0.0.0 0.255.255.255
    deny ip any 172.16.0.0 0.15.255.255
    deny ip any 192.168.0.0 0.0.255.255
    permit ip any any
```

Design notes:

- **Order matters.** The central servers sit inside 10.0.0.0/8, so their permit entries come before the private-range denies. Every other internal destination, including the TFTP server and all internal VLANs, falls under the deny.
- **Scope.** This is an internal segmentation control, not a complete Internet security policy. The final permit ip any any allows every destination outside the private ranges.
- **DHCP.** Initial lease broadcasts (destination 255.255.255.255) do not match the private-range denies and are admitted by the final permit. Unicast lease renewals to the gateway address would match the 10.0.0.0/8 deny, so deployments with long leases should add explicit bootps/bootpc entries.

8. Verification and Test Results

#	Test	Expected	Result	Evidence
1	OSPF adjacencies on GER-EDGE	Five branch edges and GER-CORE in FULL	Pass: six neighbors FULL	show ip ospf neighbor

#	Test	Expected	Result	Evidence
2	KSA-PC-IT to 10.40.20.21 (India)	Cross-branch connectivity	Pass: 4/4 replies, 0% loss, average 13 ms	ping
3	KSA-PC-IT to 10.10.70.10 (DNS)	Central server reachable	Pass: 4/4 replies, 0% loss, average 6 ms	ping
4	EGY-PC-IT to http://www.itskills.com	Name resolves, site loads	Pass: home page displayed	Web browser
5	VLAN database on DXB-CORE	Nine VLANs active	Pass	show vlan brief
6	Core-to-access trunk on DXB-CORE	802.1Q, native VLAN 99, VLANs 10–80 and 99 allowed	Pass	show interfaces trunk
7	SVIs and transit link on EGY-CORE	SVIs for VLANs 10–80 and 99, plus Gi0/1, up/up	Pass	show ip interface brief
8	Guest ACL on DXB-CORE	Nine entries as designed	Pass	show access-lists
9	Guest PC to local gateway	Ping succeeds	Pass at all six sites	Functional test
10	Guest PC to private internal host	Blocked by ACL	Pass at all six sites	Functional test
11	Guest PC to company website (HTTP)	Website loads	Pass at all six sites	Functional test

Path analysis. Packet Tracer PCs start with a TTL of 128. The reply TTL of 123 on the KSA-to-India ping means five routed hops (KSA-CORE, KSA-EDGE, GER-EDGE, IND-EDGE, IND-CORE), which confirms that cross-branch traffic transits the HQ hub as designed. The reply TTL of 124 to the DNS server matches four hops (KSA-CORE, KSA-EDGE, GER-EDGE, GER-CORE).

9. Operations Reference

Command	Purpose
`show ip ospf neighbor`	Check OSPF adjacency state and neighbor addresses
`show ip route ospf`	List routes learned through OSPF
`show ip interface brief`	Check interface and SVI status and addressing
`show vlan brief`	Confirm the VLAN database
`show interfaces trunk`	Confirm trunk mode, native VLAN and allowed VLANs
`show access-lists SITE-GUEST-IN`	Review Guest ACL entries and match counters
`show ip dhcp binding`	Review active DHCP leases on a core switch

10. Future Enhancements

The current design meets the project objectives. The following steps would extend it toward a production-grade deployment:

- **Management security:** enforce SSH-only remote access with local or centralized AAA, an enable secret and login banners.
- **Routing protection:** enable OSPF authentication on the WAN links so that only trusted routers can form adjacencies.
- **Access-layer protection:** add port security, DHCP snooping and dynamic ARP inspection on the access switches.
- **Resilience:** add a second WAN path and redundant core switching with first-hop redundancy (HSRP or VRRP) to remove the HQ single point of failure.

11. Conclusion

The IT Skills network delivers a six-site enterprise design with consistent addressing, VLAN segmentation, multilayer inter-VLAN routing, OSPF area 0 routing across a hub-and-spoke WAN, centralized DNS, web and TFTP services, and Guest VLAN isolation through extended ACLs. Verification shows all six OSPF neighbors of GER-EDGE in FULL state, successful cross-branch and central-service connectivity, and Guest clients blocked from private internal networks while retaining DNS and website access.